

LESSON 1: VENTRICLES, COVERINGS, AND VASCULAR SUPPLY OF THE BRAIN

VENTRICULAR SYSTEM

- Communicating system of cavities that are **lined with Ependyma** and **filled with Cerebrospinal Fluid** (to provide nutrients and protection).

LATERAL VENTRICLE - largest of the ventricles.

- **2 central portions** (body and atrium)
- **3 extension** (horns)
- **CHOROID PLEXUS** - **produces csf**; fringe-like vascular process of pia mater containing **capillaries of the choroid arteries**.
- **TELA CHOROIDEA** - attachment of the plexus to adjacent brain structures.
- **INTERVENTRICULAR FORAMEN (FORAMEN OF MONRO)** - communication between **lateral and third ventricle**.

THIRD VENTRICLE - narrow vertical cleft between the 2 halves of diencephalon.

- **OPTIC RECESS** - extension of 3rd ventricle **between lamina terminalis and optic chiasm**.
- **PINEAL RECESS** - projects into the **stalk of the pineal body**; secretes **melatonin**.

CEREBRAL AQUEDACT (AQUEDACT SYLVIIUS) - narrow curved channel running from posterior 3rd ventricle into 4th; **no csf existing (choroid plexus)**.

FOURTH VENTRICLE - pyramid shaped cavity **bounded ventrally by pons and medulla oblongata**.

- **MEDIAL APERTURE (FORAMEN OF MAGENDIE)** - opening of caudal portion of roof of the ventricle.
- **LATERAL APERTURE (FORAMEN OF LUSCHKA)** - opening of lateral recess into the subarachnoid space near the flocculus of the cerebellum.

MENINGES AND SPACES

SUBDURAL SPACE - separates the **Dura mater superiorly** and **arachnoid mater inferiorly**.

- **Contains a few drops of CSF**.

SUBARACHNOID SPACE - **inferior to the arachnoid mater** and **superior to the pia mater**.

- Contains CSF and major arteries of the brain
- **Types of stroke: Ischemic and Hemorrhagic stroke**.

LEPTOMENINGES - composed by **pia and dura mater**.

PERIVASCULAR SPACE (VIRCHOW-ROBIN'S SPACE) - narrow extension of subarachnoid space along the pia mater.

- Vessels deep into the brain tissue.

DURA MATER (PACHYMENINX) - tough fibrous structure with **inner (meningeal)** and **outer (periosteal) layer**.

- Most dura's venous sinuses lie between dura layers:

- **OUTER PERIOSTEAL LAYER (skull)** - firmly attached to the inner surface of the cranial bones and sends vascular and fibrous extensions into the bone itself.
- **INNER MENINGEAL LAYER** - continuous with the spinal dura.

TENTORIUM CEREBELLI

- **Separates occipital lobes from cerebellum**; roughly transverse membrane.
- Fuses with Falx Cerebri towards midline.
- Attachments:
 - Rear and side of the skull at the **transverse sinuses**
 - At front - **petrous portion of the temporal bone** and to the **clinoid process of the sphenoid bone**.
- **TENTORIAL NOTCH (INCISURA TENTORIA)** - free, curved anterior border leaves a large opening for **passage of the upper brain stem, aqueduct, and vessels**.

FALX CEREBELLI

- **Separates 2 cerebral hemispheres**; projects between cerebellar hemispheres from the inner surface of occipital bone to form a small triangular dural septum.
- Attached to the **inner surface of the skull midplain**, from the **crista galli to the internal occipital protuberance** (becomes continuous with the **tentorium cerebelli**).
- **Dura septa** - extends down into the longitudinal fissure between the cerebral hemispheres.

DIAPHRAGMA SELLAE

- Forms an incomplete lid over the hypophysis in the sella turcica by connecting the clinoid attachments of the 2 sides of the tentorium cerebelli.
- **Where pituitary stalk passes through**.

ARACHNOID MATER

- Delicate, avascular membrane that covers the **subarachnoid space**.
- Inner surface of arachnoid is connected to pia by fine **ARACHNOID TRABECULAE**
- Closely covers the inner surface of dura mater but is separated from it by **subdural space** (contains a thin film of fluid).

ARACHNOID GRANULATIONS - sites of **absorption of the CSF**.

- Consists of many **microscopic villi**
- **Berry-like clumps** protruding into the superior sagittal sinus (vein).

SUBARACHNOID SPACE - between arachnoid and pia mater; narrow in cerebral hemispheres that becomes wider in areas at the base of the brain.

- Once it's widened, it's called **CISTERN**.
- **CISTERNA MAGNA** - from bridging of the arachnoid over space between medulla and cerebral hemispheres; continuous with **spinal subarachnoid space**.
- **PONTINE CISTERN** - ventral aspect of the pons that contains the basilar artery and some veins.

WIDE SPACES BELOW CEREBRUM THAT SEPARATES TEMPORAL LOBE:

1. **CHIASMATIC CISTERNA** - above optic chiasm
2. **SUPRASELLAR CISTERNA** - above diaphragmatic sellae
3. **INTERPENDICULAR CISTERN** - between cerebral peduncles

PIA MATER

- Thin connective tissue membrane that covers the brain that **extends to sulci and fissures**.
- Extends into the transverse cerebral fissure under the Corpus Callosum
- **Forms the Tela Choroidea of the 3rd and lateral ventricles** and combines with the ependyma + choroid vessels to form **the choroid plexus of these ventricles**.

CLINICAL CORRELATION

THE TENTORIUM - dural septum that separates the **supratentorial and infratentorial compartments** (communicates through incisura (contains midbrain)).

- mass/lesion can force brain tissue through the incisura.

TRANSTERTORIAL HERNIATION (UNCAL) - the uncus (medial temporal lobe) is pushed through the tentorium, compressing the brainstem and CN III (Oculomotor)

- **Clinical sign: Ipsilateral dilated pupil and third nerve paresis.**

SUBFALCINE HERNIATION - cingulate gyrus is displaced into or under the falx cerebri.

- **Clinical sign: Lateral displacement of midline structures.**

CONING (TONSILAR) - cerebellar tonsils are forced downward into the foramen magnum.

- **Clinical sign: Rapid brainstem compression; frequently fatal.**

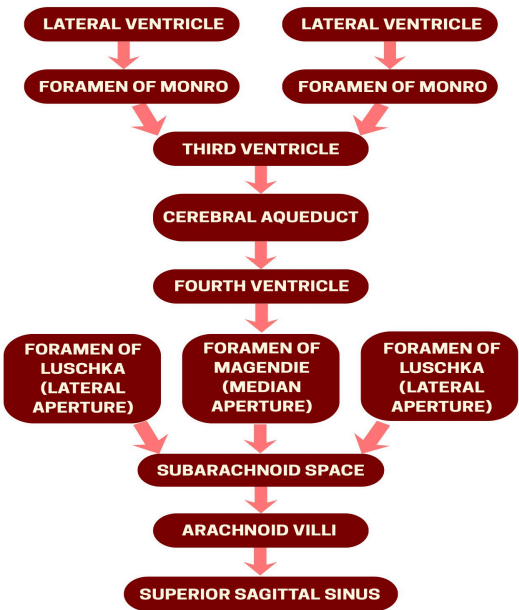
CEREBROSPINAL FLUID

- Mechanical support of the brain and **acts like a protective water jacket**.
- Controls brain excitability by **regulating ionic compositions**; carries away metabolites.
- Clear and colorless.

CSF PRESSURE

- **Normal mean - 70-180mmH2O**
- **Increase**
 - Intracranial volume (**mass/tumors or with massive infarcts that can cause swelling**)
 - blood volume (**hemorrhage**)
 - CSF volume (**Hydrocephalus**)

CSF CIRCULATION:



CSF CLINICAL CORRELATION

- Blocking the circulatory pathway of CSF leads to **hydrocephalus**.

2 types of hydrocephalus:

1. **NON-COMMUNICATING (OBSTRUCTIVE)** - most common
 - Ventricles **cannot reach subarachnoid space**
 - Block rapidly leads to **dilatation of one or more ventricles**.
 - **gyri may flatten against the skull**; can enlarge if the skull is pliable.
- **CEREBRAL AQUEDUCT** - most common site of obstruction
2. **COMMUNICATING HYDROCEPHALUS** - obstruction in **arachnoid space**
 - Can be a result of prior bleeding/meningitis
 - If intracranial pressure is raised because of CSF, the central canal of the spinal cord may dilate.
 - Spaces filled by CSF are **uniformly enlarged** without increase in intracranial pressure.
- **NORMAL PRESSURE HYDROCEPHALUS** - may be accompanied by a **gait disorder, incontinence, and dementia** (in elderly).

BARRIERS IN THE NERVOUS SYSTEM

BLOOD BRAIN BARRIER

- **Formed by:**

- **BLOOD-CSF BARRIER** - 60% is formed by active transport from blood vessels in the choroid plexus.
 - **Epithelial cells of plexus** - forms a continuous layer that **selectively permits** the passage of some substance (joined by tight junctions).
- **VASCULAR-ENDOTHELIAL BARRIER** - for promotion of exchange of oxygen, carbon dioxide, amino acids, and sugars between blood and brain.
 - Maintained by tight junctions in between.
- **ARACHNOID BARRIER** - substances diffusing out of dural vessels do not enter the CSF of subarachnoidal space.

EPENDYMA - produces and circulates CSF.

- Continuous with the epithelium of the choroid plexus.

SKULL

- Rigid in adults; pliable in infants
- Surrounds the brain and meninges completely and forms a strong mechanical protection.
 - *In adults, the volume of the brain can increase beyond the capacity of the intact cranium as a result of swelling after injury.*

SKULL (BASAL VIEW)

- **ANTERIOR PORTION OF THE BASE OF THE SKULL**
 - Hard palate
 - Below level of remainder of the inferior skull surface
- **POST PORTION OF SKULL** - Choanae
- **LATERAL TO CHOANAE** - Pterygoid plates
- **FORAMEN OVALE** - base of the pterygoid plates.
 - 3rd branch of trigeminal nerve
 - Accessory meningeal artery
 - Superficial petrosal nerve
- **FORAMEN SPINOSUM** - middle meningeal vessels; posterior to foramen ovale.
- **STYLOMASTOID FORAMEN** - base of styloid process; facial nerve exits.
- **FORAMEN LACERUM** - large irregular aperture at the base of the medial pterygoid plate.
 - Where the Internal Carotid Artery (ICA) crosses.
- **SULCUS TUBAE AUDITIVE** - contains cartilaginous part of auditory tube.
 - Forms **bony part of the auditory tube**.
- **JUGULAR FORAMEN** - posterior to carotid canal
 - **ANTERIOR** - internal petrosal sinus
 - **MIDDLE** - glossopharyngeal, vagus, and spinal accessory nerves.

- **POSTERIOR** - sigmoid sinus and meningeal branches from occipital and ascending pharyngeal arteries.
- **FORAMEN MAGNUM** - posterior to the basilar portion of the occipital lobe.
 - Medulla, spinal accessory nerves, vertebral arteries, ant. and post. spinal arteries
 - Bounded posteriorly by the **Occipital Condyles**.
- **CONDYLOID FOSSA** - perforated on one or both slides by the posterior condyloid canal; transmit emissary vein from the transverse sinus.
- **ANTERIOR CONDYLAR CANAL (HYPOGLOSSAL CANAL)** - transmits hypoglossal nerve and a meningeal artery.

INFERIOR OF THE SKULL - CALVARIA

- Concave
- With depressions for the convolutions of the cerebrum
- Furrows for the branches of the meningeal vessels
- **SUPERIOR SAGITTAL SINUS** - longitudinal groove, narrow anteriorly and posteriorly wide along the midline.

INFERIOR OF THE SKULL - FLOOR

- Divided into anterior, middle, and posterior

FORAMENS	STRUCTURES
CRIBRIFORM PLATE OF ETHMOID	Olfactory nerves
OPTIC FORAMEN	Optic nerve, ophthalmic artery, meninges
SUPERIOR ORBITAL FISSURE	Oculomotor, trochlear, and abducens nerves; ophthalmic divisions of trigeminal nerve; superior ophthalmic vein
FORAMEN ROTUNDUM	Maxillary division of trigeminal nerve, small artery, and vein
FORAMEN OVALE	Mandibular division of trigeminal nerve and vein
FORAMEN LACERUM	ICA and sympathetic plexus
FORAMEN SPINOSUM	Middle meningeal artery and vein
INTERNAL ACOUSTIC MEATUS	Facial and vestibulocochlear nerves; internal auditory artery
JUGULAR FORAMEN	Glossopharyngeal, vagus, and spinal accessory nerves; sigmoid sinus
HYPOGLOSSAL CANAL	Hypoglossal nerve
FORAMEN MAGNUM	Medulla and meninges, spinal accessory nerve, vertebral arteries, anterior and posterior spinal arteries

LECTURE 4.1: VASCULAR SUPPLY OF THE BRAIN

ARTERIAL SUPPLY

- 18% of the total blood volume
- Loss of consciousness occurs in **less than 15secs** after blood flow to the brain has stopped; **irreparable damage occurs within 5mins.**

CIRCLE OF WILLIS - confluence of vessels that **gives rise to all of the major cerebral arteries.**

★ **PRINCIPAL ARTERIES**

- **INTERNAL CAROTID ARTERY** - enters brain from neck area; **divides common carotids and vertebral artery.**
- **VERTEBRAL ARTERY** - supplies **brain stem, cerebellum, occipital lobe, and parts of the thalamus.**
 - **Carotids** - normally supply the **remainder of the forebrain.**
- **ANTERIOR COMMUNICATING ARTERY AND ANTERIOR CEREBRAL ARTERY** - interconnects the carotid arteries.
- **POSTERIOR COMMUNICATING ARTERY** - connects the **2 posterior cerebral arteries.**

★ **VERTEBROBASILAR ARTERY**

- **BASILAR ARTERY** - 2 vertebral arteries fused after passing through the foramen magnum.
 - Terminates in the **interpeduncular cistern** in a *bifurcation* as the left and right posterior cerebral arteries.
- **POST. AND ANT. INFERIOR AND SUPERIOR CEREBELLAR ARTERIES** - several pairs of **small circumferential arteries** arise from **Basilar arteries.**
 - Post. inferior cerebellar arteries
 - Ant. inferior cerebellar arteries
 - Superior cerebellar arteries
 - Pontine branches
 - Internal auditory arteries
- **PENETRATING ARTERIES** - branches off the basilar artery and supplies the brain stem.

INTERNAL CAROTID ARTERY

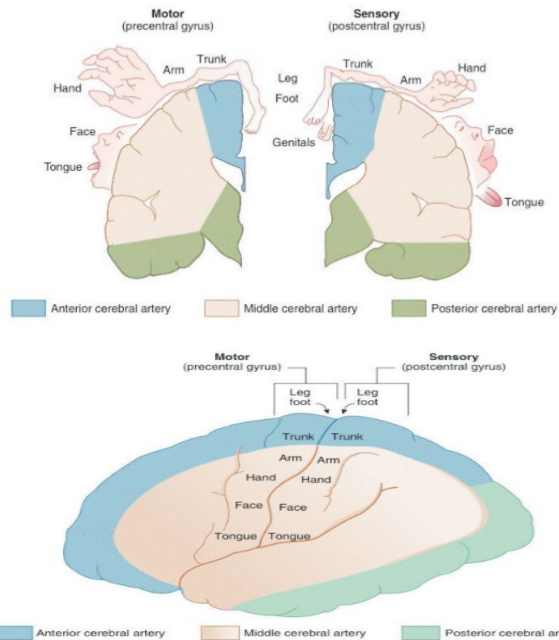
- Passes through the **carotid canal of the skull.**
 - Curves forward within the **cavernous sinus.**
 - Up and backward through the dura.
- ★ Branches into:
- **MIDDLE CEREBRAL ARTERY** - larger
 - **ANTERIOR CEREBRAL ARTERY** - smaller
- ★ **OPHTHALMIC ARTERY** - first branch of the ICA

★ **INTERNAL CAROTID ARTERY**

- **ANTERIOR COMMUNICATING ARTERIES** - connects the **2 anterior cerebral arteries.**
 - ★ Forms an **ANASTOMOSIS** between left and right hemispheres (*important when one internal carotid becomes occluded.*)
- **ANTERIOR CHOROIDAL ARTERY** - directly off the internal carotid
 - Carries blood **to the choroid plexus of the lateral ventricles** and adjacent brain structures.

★ **CORTICAL SUPPLY**

POSTERIOR CEREBRAL ARTERY	- Occipital lobe - Choroid plexuses of 3rd and lateral ventricles - Lower surface of the temporal lobe
ANTERIOR CEREBRAL ARTERY	- Medial surface of the frontal and parietal lobes, Basal ganglia, and anterior portion of the corpus callosum
MIDDLE CEREBRAL ARTERY	- Lateral surface of the frontal, parietal, and temporal portions of the brain - Basal ganglia and internal capsule



CEREBRAL BLOOD FLOW AND AUTOREGULATION

★ **Pressure at the small cerebral arteries**

- **450mmHg**
- Ensures **adequate perfusion of the cerebral capillary beds** despite changes in systemic blood pressure.
- Increased activity in one cortical area is accompanied by a **shift in blood volume** to that area.

VENOUS DRAINAGE

★ **Types of Channels:**

- Dural venous sinus
- Dural meningeal veins
- Diploic Veins

INTERNAL DRAINAGE

- ★ **GREAT CEREBRAL VEIN OF GALEN** - drains interior of the cerebrum.
 - Drainage of the **Internal Cerebral Veins and Basal Veins of Rosenthal**
- ★ **STRAIGHT SINUS** - drains from the Great Cerebral Vein of Galen and inferior sagittal sinus.
- ★ **CAVERNOUS SINUS** - receive **venous drainage** from the superior and inferior ophthalmic veins and the superficial middle cerebral vein.

CORTIAL VEINS

Venous drainage - into the nearest large vein or sinus → confluence of the sinuses → **internal jugular vein**.

- **Division of veins of the cerebral convex surface:**
 - **Superior** - upward on the hemisphere’s surface → superior sagittal sinus, **passing under any lateral lacunae**.
 - **Inferior** - end in the superficial middle cerebral vein. Inferior cerebral veins that do not end in this fashion **terminate in the transverse sinus**.

VENOUS SINUSES

INTRADURAL SINUSES - venous channels **lined by mesothelium** that lies between the inner and outer layers of the dura.

- **All sinuses drain into the internal jugular veins or pterygoid plexus.**

EMISSARY VEINS - connection of the **sinuses of the extracranial area**.

SUPERIOR SAGITTAL SINUS - between the **falx and the inside of the skull cap**.

INFERIOR SAGITTAL SINUS - in the **free edge of the falx**.

STRAIGHT SINUS - in the **seam between the falx and the tentorium**.

TRANSVERSE SINUSES - between **tentorium and its attachment on the skull cap**.

SIGMOID SINUSES - **S-curved** continuations of the transverse sinuses into the jugular veins; a **transverse and sigmoid sinus together form a lateral sinus**.

SPHENOPARIETAL SINUSES - drain the deep middle cerebral veins into the cavernous sinuses.

CAVERNOUS SINUSES (on the either side of the **sella turcica**) - receive drainage from multiple sources, including the ophthalmic and facial veins.

INFERIOR PETROSAL SINUSES - cavernous sinus → beginning of sigmoid sinus.

CEREBROVASCULAR DISORDERS

- **Most common cause of neurologic disability in adults** and the **3rd most common cause of death** in our society.
- **CNS is uniquely sensitive to ischemia** due to high metabolic rate and limited energy reserved.

- Ischemia results in rapid depletion of adenosine triphosphate (ATP) stores in the CNS.

Classification of Cerebrovascular Disease

- **Ischemic**
- **Hemorrhagic**

- **TRANSIENT ISCHEMIA/TRANSIENT ISCHEMIC ATTACK** - brief episode of neurological dysfunction resulting from an **interruption in the blood supply to the brain/eye**.

- **PENUMBRA** - surrounds the area of the infarction.
 - Neurons may have been **metabolically compromised** and are **electrically silent but are not yet dead**.
 - Neurons may be salvageable.

OCCLUSIVE CEREBROVASCULAR DISEASE

- Insufficient blood supply to portions of the brain leads to **infarction and swelling with necrosis of brain tissue**.
- Can be caused by **Atherosclerosis (most common), Thrombosis, and Embolism**.
- Extent of an infarct **depends on the presence/absence of adequate anastomotic channels**.
- Capillaries from adjacent vascular territories and corticomeningeal capillaries at the surface may reduce the size of the infarct.
- **Occlusion in the proximal of the Circle of Willis**
 - Collateral circulation through the **anterior communicating artery (ACA) and posterior communicating arteries (PCA)** may permit sufficient blood flow to prevent infarction.
- **Occlusion in the Internal Carotid Artery**
 - **Anastomotic flow in the retrograde direction** via the **ophthalmic artery**, from the external carotid artery, may provide adequate circulation thus preventing infarction.

ATHEROSCLEROSIS OF THE BRAIN

- Principal pathologic change in the arteries of the brain occurs in the **vasculature of the neck and brain**.
- **Disturbances in metabolism (fats)** - prominent associated change.
- **Hypertension** - accelerates the progression of **atherosclerosis** and is a treatable risk factor for stroke.
- **CAROTID BIFURCATION** - most common and most severe atherosclerosis locations:
 - Upper and lower parts of the basilar artery
 - Internal carotid artery at its trifurcation
 - The first third of the middle cerebral artery

NEUROSCIENCE MIDTERM

SUMALPONG, PRINCESS KAELEEN LOISSE S. OPHT 1-1

- The first part of the posterior cerebral artery.

TRANSIENT CEREBRAL ISCHEMIA

- Caused by **transient occlusion of an already narrow vessel**.
- Cause: **vasospasm**
- Full recovery in **less than 24 hours (commonly 30 minutes)**.
- *Result in reversible ischemic neurologic deficits, such as sudden vertigo or weakness, loss of cranial nerve function, or even brief loss of consciousness.*

LOCALIZATION OF THE VASCULAR LESION IN STROKE SYNDROMES

- **CAROTID ARTERY DISEASE** - often accompanied by **contralateral weakness or sensory loss**.
 - **Aphasia/apraxia** if the dominant hemisphere is involved.
 - Transient blurring/loss of vision (**amaurosis fugax**) may occur if there is retinal ischemia.
- After **occlusion of the ICA**:
 - Ischemia is often limited to the **territory of the middle cerebral artery**.
 - Weakness predominantly affects the **contralateral face and arm**.
 - The anterior and posterior cerebral artery territories are nourished via **collateral flow from the contralateral circulation** via the **anterior communicating and posterior communicating arteries**.
- **UNILATERAL OCCLUSION OF THE ANTERIOR CEREBRAL ARTERY** - weakness and sensory loss in the **contralateral leg**.
- **BILATERAL OCCLUSION OF THE ANTERIOR CEREBRAL ARTERY** - damage to **frontal lobes** → state of **akinetic mutism**: patient is indifferent and apathetic, moving little, and not speaking (even though there is no paralysis of the immobile limbs).
- **VERTEBROBASILAR ARTERY DISEASE** - presents with vertigo +
 - **Ataxia** (impaired coordination)
 - **Dysarthria** (slurred speech)
 - **Dysphasia** (impaired swallowing).
 - Nausea, and vomiting may be present
 - **Diplopia**: double vision (if oculomotor complex is involved)

HYPERTENSION HEMORRHAGE

- Caused by **chronic uncontrolled hypertension**
 - May result in the formation of **small areas of vessel distention—microaneurysm**—mostly in small arteries that arise from much larger vessels.
- Rupture of an aneurysm - **intracerebral hemorrhage**.

- **Most common sites of Intracerebral Hemorrhage** (in order of frequency)
 - **Lentiform nucleus** - specifically Putamen (**most common**).
 - **Thalamus**
 - **White matter of cerebral hemispheres**
 - **Pons**
 - **Cerebellum**

- Blood clot compresses and may destroy adjacent brain tissue.
 - Hemorrhages may compress the underlying **fourth ventricle and produce acute hydrocephalus**.

SUBARACHNOID HEMORRHAGE

- From **ruptured aneurysms or vascular malformations** (generally bleeds into the subarachnoid space).
- Vascular malformations (AVMs) often occur in **younger people** and are found on the brain surface.
- May be **congenital (berry aneurysm)** or the result of **infection**.
- Most frequent cause:
 - **Tearing of the bridging veins between brain surface and dural sinus**.
 - Can occur as a result of a relatively **minor trauma**.
- **VASCULAR SPASM** - complication that can lead to infarct.
- **CONGENITAL BERRY** - seen most frequently in the Circle of Willis or in the middle cerebral trifurcation.
 - **Common at sites of arterial branching**
 - Infrequently in vessels of the posterior fossa.

EPIDURAL HEMORRHAGE

- Most frequent cause:
 - Bleeding from a **torn meningeal vessel** (usually an artery) may lead to an **extradural** (outside the dura) accumulation of blood that can rapidly compress the brain, progressing to herniation or death.
- **Fracture of the skull** can cause this type of epidural/extradural hemorrhage.
- There is often **pulsating exophthalmos**
- **Interventional methods** - inserting a balloon/instrumentation into the shunt via a catheter or surgery may correct the problem.

ARTERIOVENOUS MALFORMATION SHUNTS

- **Cerebral arteries and veins form abnormal tangles/webs** (can occur as developmental anomalies).
- Some AVMs are clinically silent

- Trauma can also cause the rupture of adjacent vessels.

LECTURE 5: Diencephalon, Reticular System, and Limbic System

Diencephalon

- Thalamus
- Hypothalamus
- Subthalamus
- Epithalamus

3RD VENTRICLE - lies between halves of diencephalon.

HYPOTHALAMIC SULCUS - separates thalamus dorsally and the hypothalamus and subthalamic inferiorly.

Thalamus

- Thought to be the crucial structure for the perception of some types of sensation.

Landmarks:

- DORSAL THALAMUS - large, ovoid, gray mass of nuclei present in each half of the brain
- POSTERIOR - pulvinar
- ROSTRAL THALAMUS - anterior thalamic tubercle

White Matter

1. THALAMIC RADIATIONS - fiber bundles from the lateral surface of thalamus and terminate in the cerebral cortex.
2. EXTERNAL MEDULLARY LAMINA - layer of myelinated fibers on the lateral surface of the thalamus close to the internal capsule.
3. INTERNAL MEDULLARY LAMINA - vertical sheet of white matter that bifurcates in its anterior portion
 - Divides gray matter into lateral, medial, and anterior portions.

5 Groups of Thalamic Nuclei

1. Anterior Nuclear Group - forms anterior tubercle of thalamus; bordered by the limbs of the internal lamina.
 - Receives fibers from mamillary bodies via mamillothalamic tract and projects to cingulate cortex of cerebellum.
2. Nuclei of the Midline - connects with hypothalamus and central periaqueductal gray matter.
 - Centromedian sulcus - connects with the cerebellum and corpus callosum.
3. Medial Nuclei - include the intralaminar nuclei and dorsomedial nucleus that projects to the frontal cortex.
4. Lateral Nuclear Mass - constitutes a large part of the thalamus anterior to the pulvinar between internal and external medullary laminae.
 - RETICULAR NUCLEUS - between the external medullary lamina and the internal capsule.

- VENTRAL ANTERIOR NUCLEUS (VA) - connects with the corpus striatum.
- VENTRAL LATERAL NUCLEUS (VL) - projects to cerebral motor cortex.
- DORSOLATERAL NUCLEUS - projects to the parietal cortex.
- VENTRAL POSTERIOR (VENTRAL BASAL) GROUP - projects to the postcentral gyrus.
 - Receives fibers from the medial lemniscus and the spinothalamic and trigeminal tracts.
 - Divided into:
 - VENTRAL POSTEROLATERAL (VPL) - relays sensory input from the body.
 - Project information via internal capsule to the sensory cortex of the cerebral hemisphere.
 - VENTRAL POSTEROMEDIAL (VPM) - relays sensory input from the face.

5. Posterior Nuclei

- PULVINAR NUCLEUS - connects with the parietal and temporal cortices.
- MEDIAL GENICULATE NUCLEUS - receives acoustic fibers from the lateral lemniscus and inferior colliculus.
 - Projects fibers via the acoustic radiation to the temporal cortex.
- LATERAL GENICULATE NUCLEUS - major way station along the visual pathway.
 - Receives most of the fibers of the optic tract and projects via geniculocalcarine radiation to the visual cortex around calcarine fissure.
 - Geniculate nuclei/bodies - oval elevations below the posterior end of the thalamus.

5 Functional Divisions of Thalamus

1. SENSORY - relaying and modifying sensory signals from the body, face, retina, cochlea, and taste receptors.
2. MOTOR - also called as MOTOR RELAY NUCLEI
3. LIMBIC - responsible for emotions
4. MULTIMODAL NUCLEI - have connections with association areas in parietal lobe

5. **INTRALAMINAR AND RETICULAR NUCLEI** - projections are not known in detail; for consciousness

SENSORY	Lateral geniculate Medial Geniculate Ventral Posterolateral Ventral Posteromedial
MOTOR	Ventral anterior Ventral lateral
LIMBIC	Anterior Dorsomedial
MULTIMODAL	Pulvinar Lateral Posterior (Posterolateral) Lateral Dorsal (Dorsolateral)
INTRALAMINAR	Reticular Centrum medianum Intralaminar

HYPOTHALAMUS

- Serves number of autonomic, appetitive, and regulatory functions
- **Release hormones in response to any of our body's deficits**
- External landmarks:
 - **Optic Chiasm**
 - **Tuber Cinerum**
 - **Mamillary bodies**
- Divided into:
 - **Anterior portion**
 - **Chiasmatic region**, including the lamina terminalis
 - **Central hypothalamus**, including tuber cinereum and the infundibulum
 - Posterior portion
 - Mamillary area
- Left and right sides of the hypothalamus each have:
 - **Medial hypothalamic area**
 - **Lateral hypothalamic area**

MEDIAL HYPOTHALAMIC NUCLEI

1. **Tuberal Portion** - ventromedial, dorsomedial, and arcuate nuclei
2. **Supraoptic Portion** - supraoptic, suprachiasmatic, and paraventricular nuclei
3. **Mamillary Portion** - posterior nucleus and several mamillary nuclei

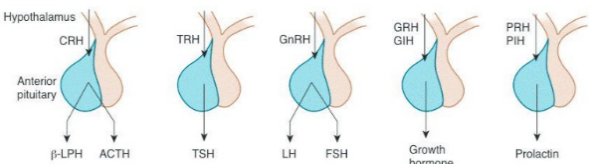
PREOPTIC AREA - lies anterior to the hypothalamus, between the optic chiasm and the anterior commissure.

Hypothalamic Afferent and Efferent Connections

ENDOCRINE

- **CRH (Corticotropin Releasing Hormone)** - send signal toward anterior pituitary gland → ACTH → Adrenal Gland → Cortisol
- **TRH (Thyrotropin Releasing Hormone)** - send signal to APG → TSH → Thyroid → T4, T3

- **GnRH (Gonadotropin Releasing Hormone)** - send signal to APG → LH/FSH ovaries: estrogen and egg cell maturation; testis: testosterone and sperm cell production
- **GRH (Growth Hormone Releasing Hormone)** - send signal to APG → GH (Growth Hormone)
- **PRH (Prolactin Releasing Hormone)** - send signal to APG → PRL - Milk production



EATING

- **LATERAL** - active feeding center; evokes eating behavior
 - Damage leads to **Anorexia** (loss of appetite) and severe loss of body weight
- **VENTROMEDIAL** - stops hunger and inhibits feeding center when high blood glucose level is reached after food intake. (sends signal kapag busog na)
 - Lesions lead to **hyperphagia** (overeating) and obesity.

AUTONOMIC FUNCTION

- **POSTEROLATERAL AND DORSOMEDIAL** - sympathetic (catecholamine) activating region
- **ANTERIOR** - parasympathetic activating region

BODY TEMPERATURE

- When stimulated evoke autonomic responses that result in loss, conservation, or production of body heat.
- **Hypothalamic Set point** - usually **37 degrees Celsius**

WATER BALANCE

- Regulates vasopressin (ADH) secretion via hypothalamic osmoreceptors near the supraoptic nucleus (thirst center).
 - Pain, stress, and certain emotional states also stimulate vasopressin secretion.

- **Vasopressin/Anti Diuretic ("Anti Ihi") Hormone** stored Posterior Pituitary Gland (activated by **osmoreceptors**) - stops flow of urine (**where sodium goes, water follows**).
- **Blood Osmolarity** - stimulates osmoreceptors; **number one trigger to release ADH**
- **DIABETES INSIPIDUS** - lack of secretion of vasopressin characterized by:

- **POLYURIA** - increased urine excretion
- **POLYDIPSIA** - increased thirst

ANTERIOR PITUITARY FUNCTION

- Controls by **releasing or inhibiting hormones via pituitary portal vessels**, directly affecting its

secretions and indirectly regulating other endocrine glands, including those involved in reproduction, thyroid and adrenal activity, and growth.

CIRCADIAN RHYTHM

- The body clock
- **SUPRACHIASMATIC GROUP** - functions as an **intrinsic clock**.
 - Appear to keep the rest of the brain on a day-night cycle
- **RETINOSUPRACHIASMATIC PATHWAY** - carries information about the **light intensity and can entertain the suprachiasmatic clock in order to synchronize its activity with environmental events**.

EXPRESSION OF EMOTION

- Involved in the expression of rage, fear, aversion, sexual behavior and pleasure.

FUNCTION	KEY INPUT	HYPOTHALAMIC AREA
Temperature	Thermoreceptors	Anterior (heat), Posterior (cold)
Catecholamines	Limbic stimuli	Dorsomedial, Posterior
Vasopressin (ADH)	Osmoreceptors	Supraoptic, Paraventricular
Oxytocin	Touch (breast, uterus, genitals)	Supraoptic, Paraventricular
TSH (via TRH)	Temperature, others	Dorsomedial
ACTH (via CRH)	Stress, limbic input	Paraventricular
FSH/LH	Estrogen, sensory input	Preoptic area
Prolactin	Breast touch	Arcuate nucleus
GH	Unknown	Periventricular, Arcuate
Thirst	Osmoreceptors	Lateral hypothalamus
Hunger	Glucose sensing	Ventromedial (satiety), Lateral (hunger)
Sexual behavior	Sex hormones	Anterior ventral hypothalamus
Fear/Rage	Sensory input	Limbic system, Hypothalamus
Circadian rhythms	Retina	Suprachiasmatic nuclei

HYPOTHALAMUS - CLINICAL CORRELATION

- **Tumors** - most common
- **Glioma, Harmatoma, Germinoma** - indirectly affected by hypothalamus
- **Pituitary Adenoma, Craniopharyngioma, Thalamic Glioma** - directly affected by hypothalamus
- **Bilateral Lesion of Lateral Hypothalamus and Reticular Component** - somnolence or coma
- **Vasopressin/Anti-Diuretic Hormone Deficiency**
 - **Diabetes Insipidus** - polydipsia and polyuria
 - **Causes:** damage to the hypothalamus because of neoplastic invasion, trauma, or vascular or infectious lesions
- **SIADH (Syndrome of Inappropriate Secretion of Antidiuretic Hormone)** - inappropriate secretion of vasopressin
 - Characterized by:
 - Hyponatremia with low plasma osmolality
 - High urinary sodium
 - No volume depletion
 - Normal kidney, liver, and adrenal function
 - Caused by:
 - Intracranial Trauma
 - Tumors
 - CNS Infections

- Inappropriate production of neoplastic cells (Small Cell Lung Cancer)

SUBTHALAMUS (MOTOR FUNCTIONS)

- Zone of brain tissue that lies between the dorsal thalamus and the tegmentum of the midbrain
- **Medial and Rostral** - Hypothalamus
- **Lateral** - Internal Capsule
- **Subthalamic nucleus/Body of Luys** - dorsolateral to the upper end of the substantia nigra

SUBTHALAMUS - FIBER CONNECTIONS

- Receives fibers from the globus pallidus and projects back to it.
- **Fields of Forel** - anterior to the red nucleus; contain cells that may be a rostral extension of reticular nuclei.

SUBTHALAMUS - Clinical Correlation

- **Hemiballismus:** motor disorder that affects one side of the body, causing coarse flailing of the arm or leg.
 - Causes:
 - **Tumor in Pineal region** - inability to move the eyes in the vertical plane (*Parinaud’s syndrome*).
 - **Germinoma** - causes precocious sexual development.
 - **Posterior commissure lesion** → loss of consensual light reflex.

EPITHALAMUS

- **Secretion of melatonin by the pineal gland**
- **HABENULAR TRIGONE**
 - Function: unknown
 - **Habenular nuclei** - receives fibers from the stria medullaris thalami and are joined via the habenular commissure.
 - **Habenulointerpenduncular tract** - extends from habenular nucleus to the interpeduncular nucleus in the midbrain.

CIRCUMVENTRICULAR ORGANS - in or near wall of 3rd ventricle, aqueduct, and 4th ventricle

- **CSF composition**
- **Hormone secretion into ventricles**
- **Maintains normal CSF pressure**

RETICULAR FORMATION (RETICULAR ACTIVATING SYSTEM)

- Plays a crucial role in **maintaining behavioral, arousal, and consciousness**.

NEUROSCIENCE MIDTERM

SUMALPONG, PRINCESS KAELEEN LOISSE S. OPHT 1-1

- Sends ascending projections into the cortex; descending axons then pass to the spinal cord in the reticulospinal tract.
- Characterized by **loosely packed, variable cells within a dense network of dendrites and axons**.
- Most of the neurons are **Serotonergic** (uses Serotonin as neurotransmitter)

Network of interconnected circuit of neurons:

- Brainstem tegmentum
- Lateral hypothalamus
- Medial intralaminar and reticular thalamic nuclei

Descending reticulospinal tract:

- Modulates spinal reflexes
- Regulates sensory input at spinal synapses
- Influences autonomic spinal activity

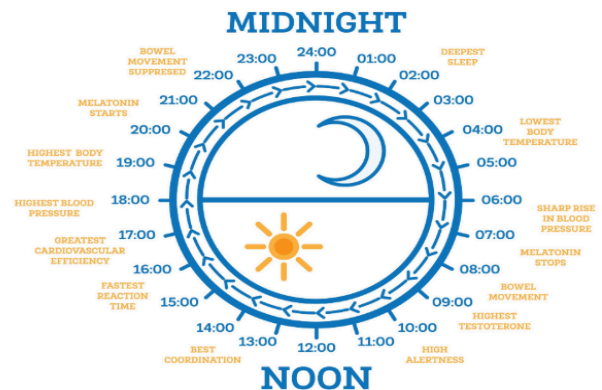
RETICULAR FORMATION FUNCTIONS:

1. **AROUSAL** - regulation of arousal and level of consciousness (generalized function of the reticular system).
 - Lesion - **anesthesia**
2. **CONSCIOUSNESS** - Cortical regions → generalized arousal when stimulated "**Binding**" of activity across cortical areas supports conscious perception and recognition.
 - **Lesions in mesencephalic reticular formation - impairs arousal.**
 - **Lesions:**
 - **Syncope (Fainting)** - brief duration and sudden onset **loss of consciousness**.
 - **Coma** (Intracranial and Externacranial (most common: **hypoglycemia**)) - prolonged and profound consciousness, **unresponsive, cannot be aroused**.
 - **Intracranial coma results from structural damage** (trauma, stroke, or tumors) that physically disrupts the brainstem or both cerebral hemispheres, often presenting with **asymmetric neurological signs**. Extracranial coma is caused by **systemic metabolic failures** (hypoxia, toxins, or organ failure) that diffusely suppress neuronal function, typically leaving the brain's physical structure intact and reflexes **symmetric**.
 - **Stupor and Obtundation** - patient is disoriented and inattentive and may be sleepy.
 - **Glasgow Coma Scale** - practical method of assessing **changes in level of consciousness**, based on eye opening, motor, and verbal responses (EMV).

3. SLEEP

- **Periodicity** - periods of sleep and waking
 - Regulated by **reticular formation, hypothalamus, and brain stem**.

CIRCADIAN RHYTHMS



- Nerve cells of the pons begin to discharge just **before the onset of sleep**.
- **Lesions of pons** - produce a state of hyperalertness and much less sleep than normal (forward the CN V).
- **Sleep stages:**
 - **SLOW WAVE SLEEP**
 - **Stage 1** - easy arousal
 - **Stage 2-4** - progressively deeper; **electroencephalograph pattern becomes more synchronized**.
 - **Stage 4 - deepest stage**; bp, pulse rate, respiratory rate and **amount of oxygen consumed by the brain are very low**.
 - **RAPID EYE MOVEMENT (REM) SLEEP** - sudden appearance of an **asynchronous pattern on electroencephalograms**.
 - **Hard to awaken**; show striking loss of muscle tone in the limbs; **vivid visual imagery and complex dreams**.
 - Triggered by **neurons in the dorsal midbrain and pontine tegmentum**.
 - Suppressed by dopa or monoamine oxidase inhibitors (increases norepinephrine concentration in the brain).
 - **PARADOXIC REM SLEEP** - second secretion (norepinephrine), produced by **locus ceruleus**, supplants the raphe secretion.
 - Effects **resemble normal wakefulness**.
- **MIDLINE RAPHE OF SYSTEM OF THE PONS** - main center responsible for bringing on sleep.

- Modifies many of the effects of the reticular activating system.
- **Destruction of the Rostral Nucleus of the Pons** - abolishes REM sleep; usually without affecting slow-wave sleep/arousal.
- **Lesions in the Raphe Nuclei** - Prolonged wakefulness.

RETICULAR FORMATION: CLINICAL CORRELATION

SOMNAMBULISM - sleepwalking (walk with their eyes open and avoid obstacles but cannot recall the episode when they awakened).

NOCTURNAL ENURESIS - bed wetting

HYPERSOMNIA - excessive daytime sleepiness

APNEA - temporary cessation of breathing

- Affected px are **obese middle-aged men who snore loudly**.
- Functional obstruction of the **oropharyngeal airway** during sleep has been implicated as a cause. (may be relieved by **tracheostomy**).
- **Periodic cessation of breathing during sleep may lead to daytime somnolence**.

NARCOLEPSY - chronic clinical syndrome characterized by **intermittent episodes of uncontrollable sleep**.

- **Cataplexy** - sudden transient loss of muscle tone.
- **Sleep Paralysis** - inability to move in the interval between sleep and arousal.
- **Hypnagogic Hallucinations** - Hallucinations occurring at the onset of sleep.

- Collectively known as **corpora quadrigemina** or the **quadrigenina plate**.

INTERNAL STRUCTURAL COMPONENTS
DESCENDING AND ASCENDING TRACTS

- All descending tracts terminating in the spinal cord **pass through the brain stem**
- All ascending tracts (e.g., spinothalamic tracts) reaching the brain stem or cerebral cortex **pass through part or all of this region**
- The brain stem - vital conduit or relay station for many longitudinal pathways.

TABLE 7–1 Major Ascending and Descending Pathways in the Brain Stem.

Ascending	Descending
Medial lemniscus	Corticospinal tract
Spinothalamic tract	Corticonuclear tract
Trigeminal lemniscus	Corticopontine fibers
Lateral lemniscus	Rubrospinal tract
Reticular system fibers	Tectospinal tract
Medial longitudinal fasciculus	Medial longitudinal fasciculus
Inferior cerebellar peduncle	Vestibulospinal tract
Superior cerebellar peduncle	Reticulospinal tract
Secondary vestibular fibers	Central tegmental tract
Secondary gustatory fibers	Descending tract of nerve V

CRANIAL NERVE NUCLEI

- All cranial nerves are located in the brainstem **EXCEPT** Olfactory (CN I) and Optic (CN II).

RETICULAR FORMATION - Areas in the tegmentum are involved in the control of respiration, cardiovascular functions, and states of consciousness, sleep, and alertness.

MONOAMINERGIC PATHWAYS

- **Serotonergic** - From the **raphe nuclei**
- **Noradrenergic** - In the lateral reticular formation and extensive efferents from the **locus coeruleus**
- **Dopaminergic** - From the **basal midbrain to the basal ganglia**

CRANIAL NERVE NUCLEI CLASSIFICATIONS

MOTOR COMPONENTS (3 type of basal plate derivatives)

1. **General somatic efferent (SE or GSE):** Striated muscles; tongue and eye movements (Hypoglossal XII, Oculomotor III, Trochlear IV, Abducens VI)
2. **Branchial efferent (BE) / Special visceral efferents (SVE):** Chewing, facial expressions, swallowing, vocal sounds, head-turning (Masticatory V, Facial VII, Ambiguus IX, X, XI, Spinal accessory XI)
3. **General visceral efferent (VE or GVE) / Parasympathetic:** Smooth muscles and glands in head, neck, and torso (Edinger-Westphal III, Superior salivatory VII, Inferior salivatory IX, Dorsal motor X)

LESSON 6: THE BRAINSTEM AND CEREBELLUM

ORGANIZATION OF THE BRAINSTEM

MAIN DIVISIONS AND LANDMARKS

- **3 main divisions:**
 - Midbrain
 - Pons
 - Medulla Oblongata
- **3 internal longitudinal divisions:**
 - Tectum (Midbrain)
 - Tegmentum
 - Basis
- **External Structures:**
 - **Superior** - Extends over the pons
 - **Inferior** - Covers the open portion of the medulla
 - **Cerebellar Peduncles (3 pairs)** - Forms connections with the cerebellum (Superior, Middle, Inferior)
- **Dorsal Aspect of the Midbrain:**
 - **4 hillocks:** 2 superior colliculi and 2 inferior colliculi

SENSORY COMPONENTS (Alar-plate derivatives)

- **General somatic afferent (SA or GSA):** Sensory stimuli from skin and mucosa of the head (**Main sensory**, descending, and mesencephalic nuclei of V)
- **General visceral afferent (VA or GVA):** Sensory stimuli from viscera and taste (Solitary nucleus IX, X; Gustatory nucleus VII, IX, X)
- **Special sensory (SS):** Four vestibular and two cochlear nuclei receiving stimuli via CN VIII

Name	Nerve	Nuclei
Oculomotor	III	Oculomotor, Edinger–Westphal
Trochlear	IV	Trochlear
Trigeminal	V	Main sensory, spinal (descending), mesencephalic, motor (masticatory)
Abducens	VI	Abducens
Facial	VII	Facial, superior salivatory, gustatory (solitary)*
Vestibulocochlear	VIII	Cochlear (two nuclei), vestibular (four nuclei)
Glossopharyngeus	IX	Ambiguus, [†] inferior salivatory, solitary*
Vagus	X	Dorsal motor, ambiguus, [†] solitary*
Accessory	XI	Spinal accessory (C1–5), ambiguus [†]
Hypoglossal	XII	Hypoglossal

MEDULLA OBLONGATA

DIVISIONS

- **Caudal (Closed):** Below the lower fourth ventricle
- **Rostral (Open):** Based on the presence of the fourth ventricle

Ascending Tracts (Caudal Closed Part) - Nucleus gracilis and nucleus cuneatus carry fine touch, proprioception, and vibration sensation; they give rise to the Medial Lemniscus.

DESCENDING TRACTS

- **Descending Spinal Tract of CN V** - Convey pain, temperature, and crude touch from the face
 - Spinal nucleus V (Pars Caudalis) - first relay station from the face
 - Mandibular division - dorsally
 - Ophthalmic division - ventrally
- **Medial Longitudinal Fasciculus** - Control of gaze and head movements
- **Tectospinal Tract** - Neck and trunk movements in response to visual stimuli

CRANIAL NERVE NUCLEI IN MEDULLA

- Below the 4th ventricle
- **HYPOGLOSSAL NUCLEUS** - innervates all tongue muscles.
- **DORSAL MOTOR NUCLEUS OF X** - controls parasympathetic tone in the heart, lungs, and abdominal viscera.
- **SUPERIOR SALIVATORY NUCLEUS** - controls salivary secretion and lacrimation
- **AMBIIGUUS NUCLEUS** - controls swallowing and vocalization
- **SOLITARY NUCLEUS (NUCLEUS SOLITARIUS)** - contains terminating axons of CN VII, IX, and X.
 - Conveys information about taste and visceral sensations
 - Gustatory nucleus - rostral part

- 4 **VESTIBULAR NUCLEI** - Sup, Inf, Med, Lat
- **COCHLEAR NUCLEI** - relay nuclei for fibers that arise in the spiral ganglion of the cochlea

INFERIOR CEREBELLAR PEDUNCLE

- Formed from cuneocerebellar; dorsal spinocerebellar tracts; fibers from lateral reticular nucleus vestibular division of nerve VII, and in vestibular nuclei; olivocerebellar fibers from contralateral inferior olivary nucleus.

PEDUNCLE - stalk-like bundle of nerve fibers containing one or more axon tracts

PONS

BASIS PONTIS

- Base of the pons; contains corticospinal tracts, pontine nuclei, and pontocerebellar fibers
- **Raphe Nuclei** - Midline neurons containing serotonin; regulate arousal, sleep-wake cycle, and modulate pain

TEGMENTUM

- **LOWER PONS** - Harbors the abducens (VI), facial, superior salivatory, and gustatory (VII) nuclei
- **UPPER HALF** -

Midbrain

- **Basis:** Includes **Crus Cerebri** (massive fiber bundle) and **Substantia Nigra** (motor control; sends dopamine to striatum; degenerates in Parkinson's)
- **Tegmentum:** Includes the **Red Nucleus** (motor coordination), Trochlear nucleus (IV), Oculomotor nuclei (III), and Edinger-Westphal nucleus
- **Tectum:** Roof of the midbrain; Superior Colliculus (ocular reflexes) and Inferior Colliculus (auditory reflexes)
- **Periaqueductal Gray Mater:** Contains cells suppressing pain; used for brain-stimulating implants in chronic pain

Vascularization and Lesions

- **Vascularization:** Supplied by branches of the Vertebrobasilar System, including basilar, vertebral, and cerebellar arteries
- **Lesions:** Vascular lesions are most common.
 - **Medial Medullary Syndrome:** Involves pyramid and CN XII; causes alternating hypoglossal hemiplegia
 - **Wallenberg's Syndrome (Lateral Medullary):** Involves CN IX, X, V, and spinothalamic tract
 - **Locked-In Syndrome:** Large basal pons lesions; interrupts corticobulbar/corticospinal pathways; patient remains awake and aware but paralyzed (speech and movement lost; eye movements often spared)
 - **Weber's Syndrome:** Midbrain lesion involving CN III and cerebral peduncle;

causes alternating oculomotor
hemiplegia

Cerebellum

- **Structure:** Located behind the pons/medulla; consists of the **Vermis** (separating two hemispheres), white matter, and **Cerebellar Cortex** (folia)
- **Divisions:**
 - **Archicerebellum:** Flocculonodular system
 - **Paleocerebellum:** Propulsive movements (walking)
 - **Neocerebellum:** Coordination of fine movement
- **Clinical Correlation:**
 - **Hypotonia:** Diminished muscle tone
 - **Ataxia:** Loss of coordinated muscular contractions
 - **Dysmetria:** Inability to place an extremity at a precise point
 - **Dysdiadochokinesia:** Difficulty with rapidly alternating movements